DEHOMOGENISING A FINITELY GENERATED GRADED ALGEBRA STAYS OF FINITE TYPE, AND A FINITE MODULE STAYS FINITE

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Allowed standard prerequisites (editorial).

- Homogeneous generators of a finitely generated graded algebra. Used at the start of the counted proof (lines 13914-13916). Reduces the generating set to homogeneous elements, which is what makes the degree bookkeeping work.
- Description of elements of the dehomogenised ring. Used throughout the counted proof (lines 13916-13943). Writes an element as a sum of fractions with bounded exponents, giving finitely many generators.

Lemma 0.1. *Let R be a ring. Let S be a graded R-algebra. Let $f \in S_+$ be homogeneous. Assume that S is of finite type over R. Then*

(1) *the ring $S_{(f)}$ is of finite type over R, and*
(2) *for any finite graded S-module M the module $M_{(f)}$ is a finite $S_{(f)}$-module.*

Proof. Choose $f_1, \ldots, f_n \in S$ which generate S as an R-algebra. We may assume that each f_i is homogeneous (by decomposing each f_i into its homogeneous components). An element of $S_{(f)}$ is a sum of the form

$$\sum\nolimits_{e \deg(f) = \sum e_i \deg(f_i)} \lambda_{e_1 \ldots e_n} f_1^{e_1} \ldots f_n^{e_n} / f^e$$

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with $\lambda_{e_1 \ldots e_n} \in R$. Thus $S_{(f)}$ is generated as an R-algebra by the $f_1^{e_1} \ldots f_n^{e_n}/f^e$ with the property that $e \deg(f) = \sum e_i \deg(f_i)$. If $e_i \geq \deg(f)$ then we can write this as

$$f_1^{e_1} \ldots f_n^{e_n}/f^e = f_i^{\deg(f)}/f^{\deg(f_i)} \cdot f_1^{e_1} \ldots f_i^{e_i - \deg(f)} \ldots f_n^{e_n}/f^{e-\deg(f_i)}$$

Thus we only need the elements $f_i^{\deg(f)}/f^{\deg(f_i)}$ as well as the elements $f_1^{e_1} \ldots f_n^{e_n}/f^e$ with $e \deg(f) = \sum e_i \deg(f_i)$ and $e_i < \deg(f)$. This is a finite list and we see that (1) is true.

To see (2) suppose that M is generated by homogeneous elements $x_1, \ldots, x_m$. Then arguing as above we find that $M_{(f)}$ is generated as an $S_{(f)}$-module by the finite list of elements of the form $f_1^{e_1} \ldots f_n^{e_n} x_j/f^e$ with $e \deg(f) = \sum e_i \deg(f_i) + \deg(x_j)$ and $e_i < \deg(f)$. $\square$